

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #11
Design of Layouts
Assembly Line Balancing

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Agenda of the Day

- Designing process layouts
- Designing product layouts
 - Line Balancing

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Designing Process Layouts

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Designing Process Layout

- ❑ The main issue in design of process layouts concerns the relative positioning of the departments involved.
- ❑ Departments must be assigned to locations.
- ❑ The problem is to develop a reasonably good layout; some combinations will be more desirable than others.

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Considerations

- ❑ Some departments may benefit from [adjacent](#) locations
 - Sharing expensive tools or equipment.
- ❑ Some departments should be [separated](#)
 - A lab with delicate equipment should not be located near a department that has equipment with strong vibrations.
 - Sand blasting department and painting department.
 - Flammable materials near a furnace.

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Measures of Effectiveness

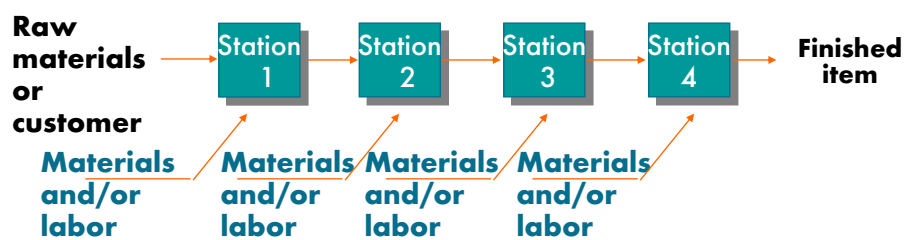
- ❑ One advantage of process layouts: satisfy a variety of processing requirements
- ❑ Customers or materials in these systems require different operations and different [sequences](#) of operations
- ❑ One of the major objectives in process layout is to minimize [transportation](#) cost, distance, or time
- ❑ This is usually accomplished by locating departments with relatively high [interdepartmental work](#) flow as close together as possible
- ❑ More information on this topic in INDR 371

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Designing Product Layouts: Line Balancing

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Production/Assembly Line



Used for Flow (Repetitive or Continuous) Processing
Example: automobile assembly lines, cafeteria serving line

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Design Product Layouts: Line Balancing

Line Balancing is the process of assigning tasks to workstations in such a way that the workstations have approximately equal time requirements.

- ❑ Tasks are grouped into manageable bundles and assigned to workstations with one or two operators
- ❑ Goal is to minimize idle time along the line, which leads to high utilization of labor and equipment
- ❑ Ideally each workstation has the same time
- ❑ However perfect balance is often impossible to achieve

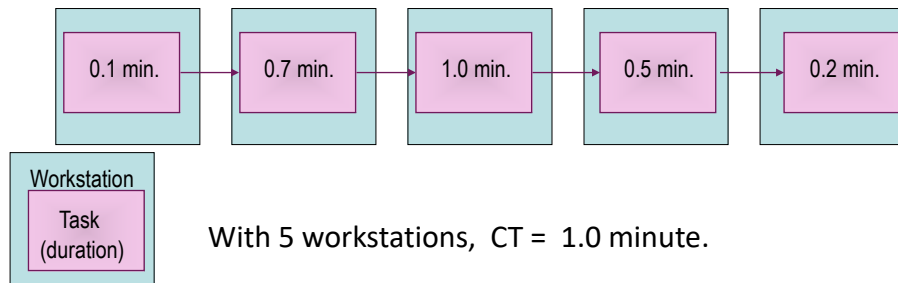
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Cycle Time

Cycle time is the maximum time allowed at each workstation to complete its set of tasks on a unit.

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Example 1: Cycle Times



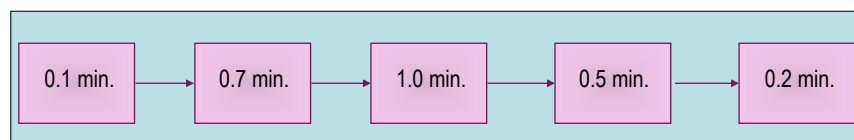
Cycle time of a system = longest processing time among all workstations.

Workstation with the longest processing time is known as “bottleneck” station and limits overall line output.

This means that the cycle time of the bottleneck station determines the cycle time of the system.

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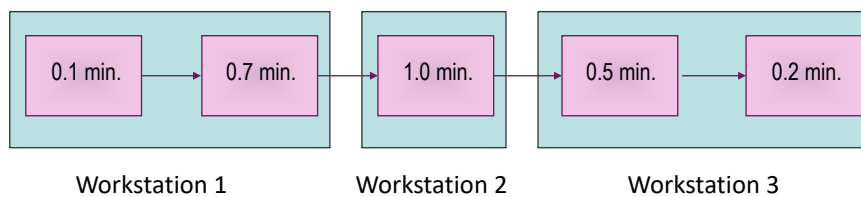
Example 1: Cycle Times



With 1 workstation, CT = 2.5 minutes.

Cycle time of workstation = total processing time of all tasks in that workstation.

With 3 workstations, can CT = 1.0 minute?



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Output Capacity

$$\text{Output capacity} = \frac{\text{OT}}{\text{CT}}$$

(Required prodn. volume)

OT = operating time per day (available time)

CT = cycle time

Example: 8 hours per day

$$\text{OT} = 8 \times 60 = 480 \text{ minutes per day}$$

$$\text{Cycle Time} = \text{CT} = 1.0 \text{ min}$$

$$\text{Output} = \text{OT}/\text{CT} = 480/1.0 = 480 \text{ units per day}$$

$$\text{Cycle Time} = \text{CT} = 2.5 \text{ min}$$

$$\text{Output} = \text{OT}/\text{CT} = 480/2.5 = 192 \text{ units per day}$$

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Cycle Time Determined by Desired Output

$$\text{CT} = \text{cycle time} = \frac{\text{OT}}{\text{D}}$$

D = Desired output rate

Example: 8 hours per day

$$\text{OT} = 8 \times 60 = 480 \text{ minutes per day}$$

$$\text{D} = 480 \text{ units per day}$$

$$\text{CT} = \text{OT}/\text{D} = 480/480 = 1.0 \text{ Minute}$$

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How to use Cycle Time?

- When we have a number of tasks to be done on different workstations in series, we need to assign tasks to workstations so that we do not exceed the cycle time.

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Theoretical Minimum Number of Stations Required

$$N_{min} = \frac{\sum t}{CT}$$

$\sum t$ = Sum of task times

N_{min} = Theoretical minimum number of workstations required

Example: 8 hours per day, desired output rate is 480 units per day

$$CT = OT/D = 480/480 = 1.0 \text{ Minute}$$

$$N_{min} = \sum t / CT = 2.5 / 1.0 = 2.5 \text{ stations}$$

$\approx 3 \text{ stations}$

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Output Capacity: When we have down time

$$\text{Output capacity} = \frac{\text{OT}}{\text{CT}}$$

OT = operating time per day

CT = cycle time

Example: OT → 8 hours/day → 8 x 60 = 480 min/day → 480-50=430 min/day (50 min down time) → 430*0.80 = 344 min/day (80% performance)

OT = 344 minutes per day

Cycle Time = **CT** = 1.0 min

Output = **OT/CT** = 344/1.0 = 344 units per day

Cycle Time = **CT** = 2.5 min

Output = **OT/CT** = 344/2.5 = 137.6 units per day

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Cycle Time Determined by Desired Output: When we have down time

$$\text{CT} = \text{cycle time} = \frac{\text{OT}}{\text{D}}$$

D = Desired output rate

Example: 8 hours per day, 50 min down time, 80% performance

OT = 344 minutes per day
((8hr/day*60min/hr-50min)*0.80)

D = 480 units per day

CT = **OT/D** = 344/480 = 0.717 Minute/Unit

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Theoretical Minimum Number of Stations Required: When we have down time

$$N_{min} = \frac{\sum t}{CT}$$

$\sum t$ = sum of task times

N_{min} = theoretical Minimum Number of Workstations Required

Example: 8 hours per day, desired output rate is 480 units per day

$$CT = OT/D = 344/480 = 0.717 \text{ minute}$$

$$N_{min} = \sum t / CT = 2.5 / 0.717 = 3.49 \text{ stations} \\ \approx 4 \text{ stations}$$

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Comparing two systems

	System without downtime	System with downtime
Output	480	344
Cycle time	1	0.717
Theoretical min. no. of machines	2.5 => 3	3.49 => 4

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Class exercise: If task times are larger than the cycle time

- A company produces toolboxes which can be used in households. The manufacturing plant works for one 8-hour shift only. The marketing department predicts that 2,000 toolboxes will be sold per day for the year 2023. The down time of the plant is estimated as 10% of the total time, whereas the plant operates with 80% performance.
- Find the desired cycle time, i.e., required number of minutes per unit, to satisfy the predicted demand. Note that you need to include the effects of down time as well as the performance.

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Class exercise

- $CT = 345.6 / 2000 = 0.173$ min/unit
- The production of the toolbox requires the assembly of different parts. The following table gives the time standards for each assembly task.
- Find the number of stations required for each assembly task.

Effective minutes:
 $(480 * 0.90) * 0.80 = 345.6$

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306				
B	0.291				
C	0.26				
D	0.356				
E	0.31				
F	0.555				
G	0.25				
H	0.415				
I	0.25				
J	0.501				

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Class exercise

- $CT=345.6/2000=0.173$ min/unit
- The production of the toolbox requires the assembly of different parts. The following table gives the time standards for each assembly task.
- Can we have decimal number of stations?

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306	1.77			
B	0.291	1.68			
C	0.26	1.50			
D	0.356	2.06			
E	0.31	1.79			
F	0.555	3.21			
G	0.25	1.45			
H	0.415	2.40			
I	0.25	1.45			
J	0.501	2.90			

No. of stations = Time std/CT;
E.g. $0.306/0.173= 1.77$ stations

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Class exercise

- $CT=345.6/2000=0.173$ min/unit
- The production of the toolbox requires the assembly of different parts. The following table gives the time standards for each assembly task.
- Given the number of stations, what is the average time spent on each machine for each assembly task?

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306	1.77	2		
B	0.291	1.68	2		
C	0.26	1.50	2		
D	0.356	2.06	3		
E	0.31	1.79	2		
F	0.555	3.21	4		
G	0.25	1.45	2		
H	0.415	2.40	3		
I	0.25	1.45	2		
J	0.501	2.90	3		

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Class exercise

- $CT=345.6/2000=0.173$ min/unit
- The production of the toolbox requires the assembly of different parts. The following table gives the time standards for each assembly process.
- What is the percentage of the time each station is busy (i.e. loaded with work)?
 Avg. time = Time std./rounded up no of m/cs;
 E.g. for A: $0.306/2=0.153$ min/unit

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306	1.77	2	0.153	
B	0.291	1.68	2	0.146	
C	0.26	1.50	2	0.130	
D	0.356	2.06	3	0.119	
E	0.31	1.79	2	0.155	
F	0.555	3.21	4	0.139	
G	0.25	1.45	2	0.125	
H	0.415	2.40	3	0.138	
I	0.25	1.45	2	0.125	
J	0.501	2.90	3	0.167	

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Class exercise

- $CT=345.6/2000=0.173$ min/unit
- The production of the toolbox requires the assembly of different parts. The following table gives the time standards for each assembly process.
 % loaded=Average time/CT;
 e.g. For A: $0.153/0.173=0.884$

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306	1.77	2	0.153	88%
B	0.291	1.68	2	0.146	84%
C	0.26	1.50	2	0.130	75%
D	0.356	2.06	3	0.119	69%
E	0.31	1.79	2	0.155	90%
F	0.555	3.21	4	0.139	80%
G	0.25	1.45	2	0.125	72%
H	0.415	2.40	3	0.138	80%
I	0.25	1.45	2	0.125	72%
J	0.501	2.90	3	0.167	97%

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Class exercise

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306	1.77	2	0.153	88%
B	0.291	1.68	2	0.146	84%
C	0.26	1.50	2	0.130	75%
D	0.356	2.06	3	0.119	69%
E	0.31	1.79	2	0.155	90%
F	0.555	3.21	4	0.139	80%
G	0.25	1.45	2	0.125	72%
H	0.415	2.40	3	0.138	80%
I	0.25	1.45	2	0.125	72%
J	0.501	2.90	3	0.167	97%

How can we use the information in this table?

What is the cycle time for the system? **Process J is called the bottleneck.**

What should we do if we want to increase the production rate?

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Class exercise

	Time standard	Theoretical # of stations	Round up	Average time	% loaded
A	0.306	1.77	2	0.153	88%
B	0.291	1.68	2	0.146	84%
C	0.26	1.50	2	0.130	75%
D	0.356	2.06	3	0.119	69%
E	0.31	1.79	2	0.155	90%
F	0.555	3.21	4	0.139	80%
G	0.25	1.45	2	0.125	72%
H	0.415	2.40	3	0.138	80%
I	0.25	1.45	2	0.125	72%
J	0.501	2.90	4	0.125	72%

What is the cycle time now?

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Class exercise

	Time standard	# of stations	Round up	Average time	% loaded
A	0.306	1.77	2	0.153	88%
B	0.291	1.68	2	0.146	84%
C	0.26	1.50	2	0.130	75%
D	0.356	2.06	3	0.119	69%
E	0.31	1.79	2	0.155	90%
F	0.555	3.21	4	0.139	80%
G	0.25	1.45	2	0.125	72%
H	0.415	2.40	3	0.138	80%
I	0.25	1.45	2	0.125	72%
J	0.501	2.90	3	0.167	97%

How else can we use the information in this table?

How about utilizations?

Can we combine two processes? e.g., C&G or C&I? How about C&D?

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Class exercise

	Time standard	# of stations	Round up	Average time	% loaded
C	0.26	1.50	2	0.130	75%
D	0.356	2.06	3	0.119	69%
C&D	0.616	3.56	4	0.154	89%

Combining C and D needs 4 workers, instead of 5 workers with separate C and D.

This improves the overall utilization, which, in turn, reduces the cost.

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Info about Midterm Exam

- ▣ Midterm exam is scheduled on 25 April 2026, Saturday during 11:45-14:45
- ▣ Detailed information will be given in due course